

UniTS Anomaly Ratio Sweep

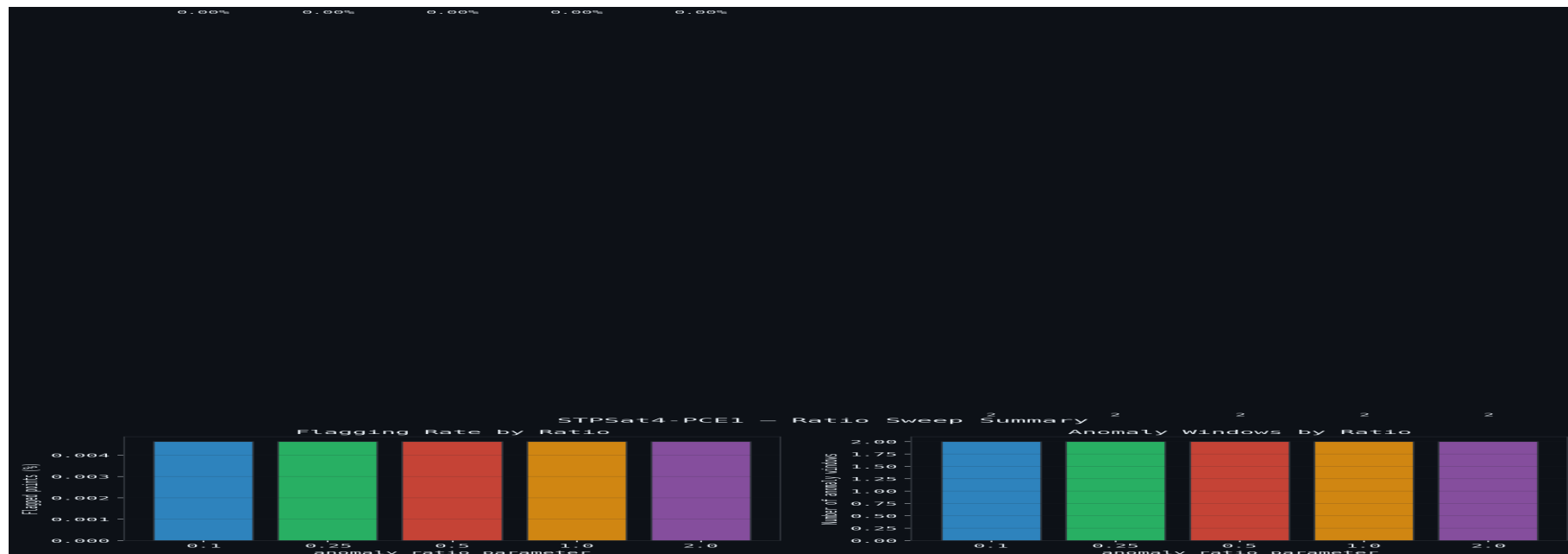
STPSat4-PCE1 | Generated 2026-03-12 10:53

Results by Ratio

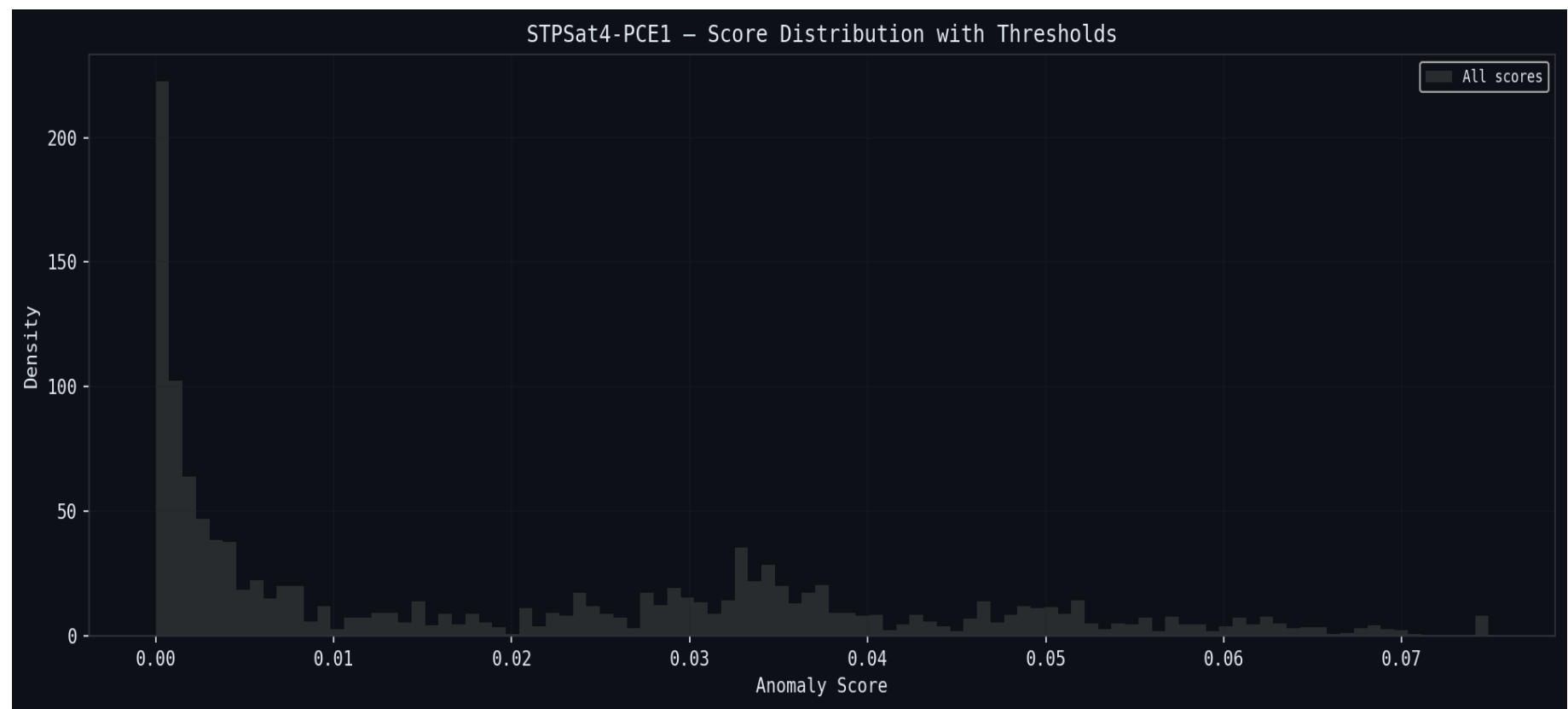
anomaly_ratio	Threshold	Flagged Points	Flag Rate	Windows
0.1	0.128594	2	0.005%	2
0.25	0.128594	2	0.005%	2
0.5	0.128594	2	0.005%	2
1.0	0.128594	2	0.005%	2
2.0	0.128594	2	0.005%	2

anomaly_ratio controls the detection threshold: UniTS flags the top ratio% of reconstruction error scores as anomalies. Lower values = stricter threshold = fewer flags. The threshold column shows the exact score cutoff for each ratio.

Flagging Rate & Window Count



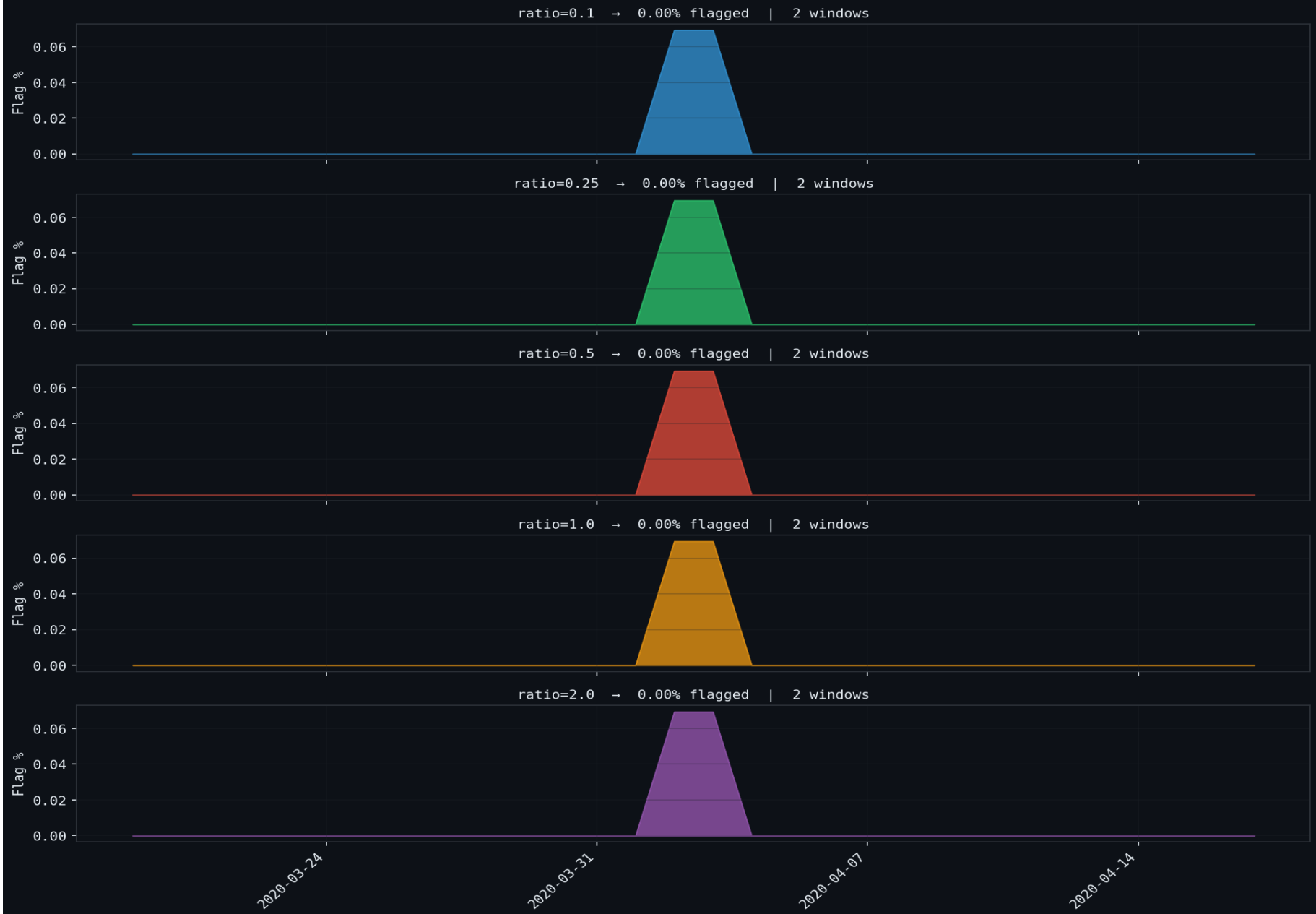
Score Distribution with Thresholds



Dashed vertical lines show where each ratio sets its threshold. Choose a ratio whose threshold sits clearly in the tail of the distribution rather than in the main body of normal scores.

Daily Flagging Rate by Ratio

STPSat4-PCE1 – Daily Flagging Rate by Ratio



A realistic anomaly rate for a nominal satellite is typically 0.1–1%. Look for a ratio that shows isolated spikes rather than sustained elevated flagging.